Solved Exercises

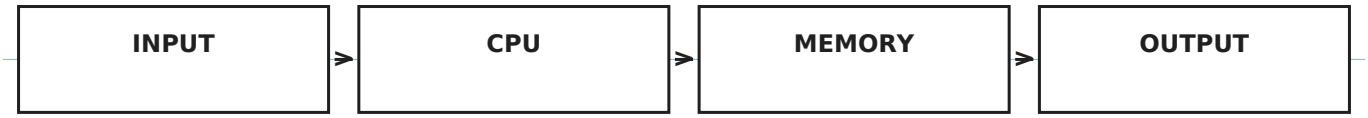
Computer Science - Class 11

Punjab Curriculum and Textbook Board (PCTB)

Solved exercises of the PCTB Class 11 Computer Science textbook (English medium; cover 2018-19, 1st edition, 7th impression, April 2018). Every chapter-end exercise is solved: fill in the blanks, MCQs, True/False and short questions. Objective answers follow the book's printed answer keys, except where a key note at the end of a chapter records a correction. Source: Class11-ComputerScience-taleem360.pdf (Taleem360 PCTB listing, 171 pages, complete book pp. 1-161 plus glossary and index).

Chapter 1: Basics of Information Technology (Book pp. 1-24, Exercise 1C pp. 22-24)

Computer System - main parts



Exercise 1C - 1. Fill in the blanks

(i) A bridge is used where the _____ type of networks are to be joined together.

Answer: **Similar**

(ii) WAN stands for _____.

Answer: **Wide Area Network**

(iii) A set of instructions given to the computer to solve a problem is called _____.

Answer: **Program**

(iv) _____ and _____ printers are non-impact printers.

Answer: **Inkjet, Laser**

(v) Trackball is popular among users of _____ computer.

Answer: **Laptop**

(vi) SVGA stands for _____.

Answer: **Super Video Graphic Array**

(vii) 1024 GB are equal to _____.

Answer: **One Tera Byte**

(viii) OMR reads _____ marks and converts them into computer-usable form.

Answer: **SAT or GRE**

(ix) LCD stands for _____.

Answer: **Liquid Crystal Display**

(x) Fax stands for _____.

Answer: **Facsimile**

(xi) Dot matrix printer is a(n) _____ type printer.

Answer: **Impact**

Exercise 1C - 2. Choose the correct option

(i) The name for the screen clarity:

a) Resolution ✓

b) Discrete

c) Pixel

d) LCD

(ii) Another word for pointer:

a) Monochrome

b) Pixel

c) Cursor ✓

d) None of the above

(iii) A device used for optical-character recognition is a:

a) Wand reader

b) Cursor

c) Pen

d) MICR reader ✓

(iv) Imaging uses what device to input data:

a) scanner ✓

b) icon

c) bar code reader

d) tablet

(v) An ink-jet printer is an example of a(n):

a) Laser printer

b) Impact printer

c) COM printer

d) Non-impact printer ✓

(vi) Soft copy refers to:

a) Screen output ✓

b) Peripheral devices

c) OCR

d) None of the above

(vii) Smallest unit of memory is:

a) Byte

b) Bit ✓

c) Character

d) Word

(viii) The printer which can print one complete line at a time is:

a) Dot matrix printer

b) Daisy wheel printer

c) Laser printer

d) Line printer ✓

(ix) The microphone converts the sound into:

a) Mechanical signals

b) Electrical signal ✓

c) Computer file

d) Software

(x) An input device, which is used for playing computer games:

a) Light Pen

b) Mouse

c) Joy Stick ✓

d) Scanner

Exercise 1C - 3. Write T for true and F for false statement

(i) The keyboard arrangement provided as standard on most keyboards is the QWERTY arrangement.

Answer: True

(ii) A picture element on the screen is called a pixel.

Answer: True

(iii) CRTs are used on portable computers.

Answer: False

(iv) Audio-output device can output only music.

Answer: False

(v) Non-impact printers are quieter than impact printers.

Answer: True

(vi) A trackball is a pointing input device almost like a mouse turned upside down.

Answer: True

(vii) The disk drives are known as I/O devices.

Answer: True

(viii) Function keys are used the same way with every software application.

Answer: False

(ix) EGA stands for Extended Graphic Adapter.

Answer: False

(x) The two basic types of plotter are the Drum plotter and the Flatbed plotter.

Answer: True

Exercise 1C - Short questions

4. What is the difference between hardware and software?

Hardware is the physical part of a computer system that you can see and touch, such as the keyboard, monitor, CPU and printer. Software is the set of programs and instructions that tell the hardware what to do; it cannot be touched. Hardware without software is useless, and software cannot run without hardware. Together they form a working computer system.

5. What is Information Technology? Describe it in detail.

Information Technology (IT) is the technology that merges computing with high-speed communication links to carry data in the form of text, sound, images and video from place to place over the global village. It allows data stored or processed on computers to be shared and exchanged almost as if users were sitting face to face. IT covers hardware, software, networks and communication systems, and it is used in education, medicine, business, research and daily life.

6. What is implementation? Describe its approaches.

Implementation is the stage of the system development life cycle in which the new system is actually put into use after testing. Its approaches are: direct implementation, in which the old system is replaced at once; parallel implementation, in which the old and new systems run together for a time; and phased implementation, in which the new system is introduced in parts. Each approach balances risk, cost and the time needed for users to adjust.

7. What is the usage of an input device? List a few of them.

An input device is used to enter data and instructions into the computer so that the CPU can process them. Common input devices are the keyboard, mouse, joystick, trackball, scanner, light pen, digital camera, microphone and optical mark reader (OMR). Each device converts a different form of data, such as typing, pointing, pictures or sound, into a form the computer can use.

8. Describe the enhanced keyboard and its segments.

The enhanced keyboard is the standard keyboard with 101 or more keys, arranged for fast and comfortable typing. Its segments are the typewriter keys (letters, digits and punctuation), the function keys (F1 to F12) used for shortcuts, the numeric keypad for quick number entry, the cursor-control and navigation keys (arrows, Home, End, Page Up, Page Down), and special keys such as Ctrl, Alt, Esc and Enter. The QWERTY arrangement is the standard layout of its letter keys.

9. Write the name of the input device that can control the cursor movement.

The mouse is the main input device used to control the movement of the cursor (pointer) on the screen. Moving the mouse on a flat surface moves the pointer, and its buttons are used to select, open and drag objects. The joystick and the trackball can also control cursor movement, but the mouse is the standard pointing device for this purpose.

10. What is the function of an output device? List a few of them.

An output device is used to present the processed result of the computer to the user in a usable form, such as on screen or on paper. Common output devices are the monitor (VDU), printer, plotter, speaker and headphone. The monitor gives soft copy output on the screen, while the printer and plotter give hard copy output on paper.

11. Describe the functioning of a Laser printer.

A laser printer is a non-impact printer that prints a whole page at a time with high speed and fine quality. A laser beam draws the image of the page on a light-sensitive drum, toner powder sticks to the charged parts of the drum, and the toner is then transferred to paper and fixed by heat. Because nothing strikes the paper, laser printing is quiet, and it is widely used for sharp text and graphics.

12. What is the usage of a plotter? Name its different kinds.

A plotter is an output device used to draw large, accurate graphics such as engineering drawings, maps, building plans and charts, which ordinary printers cannot draw well. Its different kinds are the drum plotter, in which the paper moves on a rotating drum, and the flatbed plotter, in which the paper stays flat on a table while the pen moves over it.

13. Define the term 'Operating System' in your own words.

An operating system is the system software that manages all the resources of a computer and acts as a link between the user and the hardware. It controls the CPU, memory, input and output devices and files, and it provides the platform on which application programs run. Without an operating system, the user could not run programs or use the hardware easily.

14. Describe the function of the following input devices: (i) Mouse (ii) Joystick (iii) Trackball (iv) Scanner (v) Light Pen (vi) Digital Camera.

(i) Mouse: controls the cursor and selects objects by pointing, clicking and dragging. (ii) Joystick: controls movement, mainly in games and simulators, by tilting a stick. (iii) Trackball: controls the cursor by rotating a ball held in a fixed base; it is popular with

laptop users. (iv) Scanner: copies pictures, text and documents into the computer as digital images. (v) Light Pen: points directly at the screen to select or draw. (vi) Digital Camera: captures still pictures and video directly in digital form for the computer.

15. What do you understand by 'SDLC'? Define its steps properly.

SDLC means System Development Life Cycle: the series of steps used to build a working information system. Its steps are: (1) Problem definition, stating what the system must do; (2) Feasibility study, checking whether the system is possible and worthwhile; (3) System analysis, studying the present system and the user's needs; (4) System design, planning inputs, outputs, files and programs; (5) Coding, writing the programs; (6) Testing, checking the system for errors; (7) Implementation, putting the new system into use; and (8) Maintenance, improving the system and removing problems reported by users.

Chapter 2: Information Networks (Book pp. 25-40, Exercise 2C pp. 39-40)

Exercise 2C - 1. Fill in the blanks

(i) Collection of raw facts is called _____.

Answer: **Data**

(ii) A receiver is also called _____.

Answer: **Sink**

(iii) Two forms of data transmission are _____ and _____.

Answer: **Serial, Parallel**

(iv) TCP/IP stands for _____.

Answer: **Transmission Control Protocol / Internet Protocol**

(v) Data in _____ can travel in both directions but not at the same time.

Answer: **Half Duplex Mode**

(vi) _____ ensures that data are transmitted without any error.

Answer: **Data Link Layer**

(vii) Data transmission through a medium can be either synchronous or _____.

Answer: **Asynchronous**

(viii) WAN stands for _____.

Answer: **Wide Area Network**

(ix) A _____ is a microwave station placed in outer space.

Answer: **Satellite**

(x) A router is also used as a _____ device used for interconnecting different types of networks together.

Answer: **Intermediate**

Exercise 2C - 2. Choose the correct option

(i) A LAN is a combination of:

a) Network adapter cards

b) LAN cables

c) LAN application software

d) All of the above ✓

(ii) What layer of OSI model does data compression:

a) Network

b) Presentation ✓

c) Data Link

d) Physical

(iii) Cabling on a linear bus topology can be extended using which of the following?

a) Terminator

b) Barrel connector ✓

c) Network adapter card

d) Medium attachment

(iv) The Media Access Control sub layer resides in which layer?

a) Physical

b) Data link ✓

c) Network

d) Transport

(v) FDDI is a:

a) Ring network ✓

b) Star network

c) Mesh network

d) Bus network

(vi) How many pairs of computers can simultaneously communicate on Ethernet LAN?

a) 1

b) 2

c) 3

d) Multiple ✓

(vii) One or more computers connected to a hub computer is a:

a) Ring network

b) Node

c) Information utility

d) Star network ✓

(viii) Project 802 defines standards for which layers of the OSI model?

a) Application and presentation layers

b) Physical and Data Link layers ✓

c) Transport and Network layers

d) Network and Data Link layers

(ix) Terminal is a:

a) Device to give power supply to computer

b) Point at which data enters or leaves the computer ✓

c) The last instruction in a program

d) Any input/output device

(x) Software to peruse the internet:

a) Gateway

b) EFT

c) Browser ✓

d) Teleconferencing

Exercise 2C - 3. Write T for true and F for false statement

(i) E-mail is short for electronic mail.

Answer: True

(ii) Teleprocessing allows a user to make queries of a computer 1000 miles away.

Answer: False

(iii) An Ethernet system (IEEE 802.3 protocol) uses packet switching technique.

Answer: False

(iv) ISDN modems can communicate only with other ISDN modems.

Answer: False

(v) 16 bit and 32 bit are currently the two most popular bus widths.

Answer: False

(vi) FTP, short for File Transfer Protocol, is a tool that lets users transfer files across the internet.

Answer: True

(vii) DSL modem uses the cable TV network for data transmission.

Answer: False

(viii) A WAN is usually limited to one office building.

Answer: False

(ix) A gateway connects two similar computers.

Answer: False

(x) A bus network uses a central computer as the server.

Answer: False

Exercise 2C - Short questions

4. What is the difference between LAN and WAN?

A LAN (Local Area Network) connects computers within a small area such as one building, office or campus, at high speed and low cost. A WAN (Wide Area Network) connects computers over a large area, such as cities or countries, using telephone lines, microwaves or satellites, and it is slower and more expensive than a LAN. In short, LAN is local and fast, while WAN is wide-area and long-distance.

5. What method does an Ethernet network use to control access to the network?

An Ethernet network controls access by the CSMA/CD method: Carrier Sense Multiple Access with Collision Detection. Each computer first listens to the cable; if the cable is free it sends its data, and if two computers send at the same time and a collision occurs, both stop, wait a random time, and try again. This sharing rule lets many computers use the same cable without a central controller.

6. What is Topology? Describe the types of topology.

Topology is the physical arrangement or layout by which computers are connected in a network. Its main types are: bus topology, where all computers share one central cable; ring topology, where computers are joined in a closed ring; star topology, where every computer is connected to a central hub; mesh topology, where computers are directly connected to each other; and tree topology, which combines bus and star layouts in a hierarchy.

7. What is WWW?

WWW stands for World Wide Web: the system of linked documents and resources on the internet that are viewed with a web browser. Each document, called a web page, is held on a web server and is reached by its address (URL), and pages are linked to each other by hyperlinks. The Web lets users read, search and share text, pictures, sound and video around the world.

8. Describe the OSI Model and types of layers of OSI Model.

The OSI model is a seven-layer standard that describes how data moves from one computer to another over a network. Its layers, from top to bottom, are: Application, Presentation, Session, Transport, Network, Data Link and Physical. Each layer performs one part of the job, such as addressing, routing, error control and the physical transmission of bits, so that different systems can communicate with each other.

9. Difference between Intranet and Extranet.

An intranet is a private network inside one organisation that uses internet technology but is open only to its own staff, for sharing internal information. An extranet is an extension of an intranet that also gives controlled access to selected outsiders, such as business partners or customers. So an intranet serves insiders only, while an extranet serves insiders plus trusted outsiders.

10. Difference between FTP and HTTP.

FTP (File Transfer Protocol) is used to transfer complete files between computers over the internet, such as uploading and downloading files. HTTP (Hyper Text Transfer Protocol) is used to transfer web pages, with their text, images and links, from a web server to a web browser. In short, FTP moves files, while HTTP moves web pages for viewing.

11. Define the gateway and router.

A gateway is a device or program that connects two different types of networks and

translates data between them so that they can communicate. A router is a device that connects networks of the same or similar type and forwards data packets along the best path towards their destination. The gateway translates between different systems, while the router directs traffic between networks.

12. Define the term 'Operating System' in your own words.

An operating system is the system software that manages the computer's hardware and provides the base on which all other programs run. It controls the processor, memory, storage and input/output devices, manages files, and gives the user a way to run programs. Common examples are Windows and Linux.

13. Describe the function of (i) Dedicated Server Network (ii) Peer-to-Peer Network.

(i) Dedicated Server Network: one powerful computer, the server, is set aside to provide services such as files and printing to the other computers (clients); it gives central control and security. (ii) Peer-to-Peer Network: all computers are equal and share their resources directly with each other, with no dedicated server; it is simple and cheap for small networks but harder to control and secure.

Key notes (this chapter):

- Chapter 2, MCQ (ix): the book's printed key lists option (d); the chapter's own definition of a terminal (the point at which data enters or leaves the computer) supports option (b), which is the answer given here.

Chapter 3: Data Communications (Book pp. 41-56, Exercises 3C pp. 55-56)

Exercise 3C - 1. Fill in the blanks

(i) In _____ transmission, a start bit and a stop bit frame a character byte.

Answer: **Asynchronous**

(ii) Data communication signals can be in _____ or _____ form.

Answer: **Digital, Analog**

(iii) Modem is an electrical device that converts digital signals into analog signals, which is called _____.

Answer: **Modulation**

(iv) The _____ transmission involves the concurrent flow of bits of data through separate communication lines.

Answer: **Parallel**

(v) ASCII is _____ bit(s) code.

Answer: **7 (seven)**

(vi) A television broadcast is an example of _____ transmission.

Answer: **Simplex**

(vii) In _____ transmission data is transmitted character by character.

Answer: **Serial**

(viii) The data is transmitted in both directions simultaneously on the same channel: _____.

Answer: **Full Duplex**

(ix) Fiber optic cable is better for very high speed, high-capacity data transmission than _____ cable because of the lack of attenuation and the purity of the signal.

Answer: **Coaxial**

(x) The number of frequencies that can fit on a link at one time is called _____.

Answer: **Bandwidth**

Exercise 3C - 2. Choose the correct option

(i) Microwave transmission, coaxial cables, and fiber optics are examples of:

a) Modems

b) Communication links ✓

c) Gateways

d) Ring network

(ii) Data communication requires only a:

a) Sender

b) Receiver

c) Transmission Medium

d) All of the above ✓

(iii) BIT stands for _____:

a) Binary Integer

b) Binary digit ✓

c) Binary interval

d) None of the above

(iv) Communication between a computer and a keyboard involves _____ transmission:

a) Simplex ✓

b) Half-duplex

c) Full-duplex

d) Automatic

(v) The _____ is the physical path over which a message travels:

a) Protocol

b) Medium ✓

c) Single

d) All of the above

Exercise 3C - 3. Write T for true and F for false statement

(i) An internal modem is a circuit board that can be added to the system unit of the computer.

Answer: True

(ii) In full-duplex transmission, the channel capacity is shared by both communicating devices at all times.

Answer: True

(iii) A fiber-optic cable consists of tubes of glass through which data are transmitted as pulses of light.

Answer: True

(iv) Normally, modem transmission is asynchronous.

Answer: True

(v) Transmission of signals across a communication medium is called signaling.

Answer: False

(vi) The voice channel has a bandwidth of 0-233 kHz.

Answer: False

(vii) Synchronous transmission is much faster than asynchronous.

Answer: True

Exercise 3C - Short questions

4. What is data communication? Define the basic components of a communication network.

Data communication is the transfer of data or information from one place to another through a communication channel, such as a cable or wireless link. Its five basic components are: the message (the data to be sent); the sender (the device that sends it); the receiver (the device that receives it); the communication channel (the path, such as twisted pair, coaxial cable or fiber optic); and the encoder/decoder, which converts the message into a form fit for the channel and back again.

5. Define the modem and its types.

A modem is a device that converts digital signals from a computer into analog signals for transmission over telephone lines (modulation), and converts received analog signals back into digital form (demodulation). Its types are the internal modem, fitted inside the system unit as a circuit board, and the external modem, connected outside the computer as a separate box.

6. Differentiate between Asynchronous and Synchronous transmission.

In asynchronous transmission, data are sent one character at a time, with a start bit before and a stop bit after each character, so the receiver can resynchronise on every character; it is simple but slower. In synchronous transmission, data are sent as a continuous block of characters controlled by a common clock, with no start and stop bits on each character; it is faster and is used for large volumes of data, but it needs more expensive equipment.

7. Briefly explain the various communication media.

Communication media are the paths that carry data. Guided media carry signals along a physical path: twisted pair cable (ordinary telephone wire), coaxial cable (a central copper wire with shielding, used for cable TV and networks) and fiber-optic cable (thin glass strands carrying pulses of light, very fast and free from interference). Unguided media carry signals through the air or space without a wire: microwaves, radio waves, infrared and satellites.

8. Define the Modem.

A modem (modulator-demodulator) is a communication device that lets a computer send and receive data over telephone lines. It modulates, that is, converts the computer's digital signals into analog signals for sending, and demodulates, that is, converts

incoming analog signals back into digital signals the computer can use.

9. What is bandwidth? Explain bandwidth briefly.

Bandwidth is the capacity of a communication channel: the range of frequencies it can carry, or the amount of data it can transmit in a given time. A channel with greater bandwidth can carry more data at once and at higher speed. For example, a voice telephone channel has a narrow bandwidth, while a broadband channel used for high-speed data has a much wider bandwidth.

10. Differentiate between guided media and unguided media.

Guided media, also called bounded media, guide signals along a fixed physical path, such as twisted pair, coaxial cable or fiber-optic cable; they are protected from outside interference but must be laid down. Unguided media, also called unbounded media, send signals through the air or space, such as microwaves, radio waves and satellites; they need no cabling and cover long distances, but they can suffer from interference and weather.

Chapter 4: Applications and Uses of Computers (Book pp. 57-70, Exercise 4C pp. 68-70)

Exercise 4C - 1. Fill in the blanks

(i) CAD stands for _____.

Answer: **Computer Aided Design**

(ii) Electronic banking allows individuals to obtain cash instantly from _____.

Answer: **ATM**

(iii) CAT stands for _____.

Answer: **Computerized Axial Topography**

(iv) VCR stands for _____.

Answer: **Video Cassette Recorder**

(v) A robot is an automatic programmable machine that moves and performs _____ tasks.

Answer: **Mechanical**

(vi) An electronic banking is also known as _____.

Answer: **Cyber Banking**

(vii) The cheques are read by _____ machine in computerized bank.

Answer: **MICR**

(viii) The _____ is a type of conferencing in which video cameras and microphones capture sight and sound transmission over networks.

Answer: **Video Conferencing**

(ix) Office Support Systems enable to coordinate and manage the activities of _____.

Answer: **Workgroup**

(x) FAX stands for _____.

Answer: **Facsimile**

Exercise 4C - 2. Choose the correct option

(i) CBT stands for:

a) Computer Based Trade

b) Computer Based Training ✓

c) Certificate Based Training

d) None of the above

(ii) The benefit of CAD may be summed up as:

a) Accuracy

b) Repeatability

c) Speed and flexibility of production

d) All of the above ✓

(iii) Computer at home can be used:

a) Keeping records

b) Making budgets

c) Watching Movies

d) All of the above ✓

(iv) A word processor can be used to:

a) Write Text

b) Edit Text

c) Print Text

d) All of the above ✓

(v) CAL stands for:

a) Computer Aided Learning ✓

b) Computer Assist Learning

c) Computer Added Learning

d) None of the above

(vi) Typically, an ATM can be used to:

a) Keeping records

b) Making budgets

c) Watching Movies

d) None of the above ✓

(vii) Modern computer can perform calculations or process at _____ high speed:

a) per second

b) per minute

c) nano second ✓

d) None of the above

(viii) CAT stands for:

a) Computerised Axial Topography ✓

b) Computer Axial Topography

c) Computer Aided Topography

d) None of the above

(ix) Computer based weather forecasting depends on accurate collection of data from:

a) Television

b) Weather stations ✓

c) Radar

d) Antenna

(x) MICR stands for:

a) Magic in Character Redo

b) Magnetic Ink Character Recorder

c) Magnetic Ink Character Reader ✓

d) None of the above

Exercise 4C - 3. Write T for true and F for false statement

(i) CBT is more expensive than non-CBT training.

Answer: True

(ii) Videoconferencing is an advanced form of teleconferencing.

Answer: True

(iii) The e-shop has opened for limited time period on the web sites.

Answer: False

(iv) CAL could be described as the use of information technology to assist in the teaching and learning processes.

Answer: True

(v) Bar Code Reader can be read all types of ink characters.

Answer: False

(vi) Fax machine can be inserted inside computers.

Answer: False

(vii) A robot is an automatic programmable machine.

Answer: True

(viii) A computer simulation is a special type of computer hardware.

Answer: False

(ix) An electronic banking is also known as cyber-banking.

Answer: True

(x) Modern computer can perform calculations at a second.

Answer: False

Exercise 4C - Short questions

5. Explain the term Computer Aided Manufacturing process.

Computer Aided Manufacturing (CAM) is the use of computers to control and manage the manufacturing process in a factory. The computer directs machines and robots that cut, shape, assemble and test products, following a design stored in the system. CAM gives accurate, fast and repeated production with less waste, and it is widely used in automobile, textile and electronics industries.

6. What is meant by computer simulation?

Computer simulation means using a computer to imitate a real process or system so that it can be studied without doing the real thing. The computer follows a model of the real system and shows what would happen under different conditions. Simulations are used to

train pilots, test new cars and aeroplanes, predict weather and study scientific problems that would be dangerous or expensive to test in reality.

7. What is an ATM?

An ATM (Automated Teller Machine) is a computerised banking machine that lets customers obtain cash and carry out banking services at any time, without going inside a bank. By inserting a bank card and entering a personal identification number (PIN), a customer can withdraw cash, check the account balance and get a mini statement. ATMs are the best-known face of electronic banking.

8. Explain how computer can be useful in business.

Computers help business by keeping records of sales, stock, customers and accounts quickly and accurately. They prepare bills, payrolls and reports, support electronic shopping and banking, and let offices communicate by e-mail and video conferencing. With fast and reliable information, managers can take better decisions, serve customers faster and reduce the cost of routine work.

9. Explain how computer can be useful in medical field.

In the medical field, computers keep patients' records, help doctors diagnose disease, and control laboratory tests and scanning machines such as CAT (Computerised Axial Topography) scanners, which build detailed pictures of the inside of the body. Computers also support surgery, monitor patients in intensive care, manage hospitals and pharmacies, and let doctors consult specialists far away.

10. Define the role of E-Commerce in our daily life.

E-Commerce means buying and selling goods and services over the internet. In daily life it lets people shop from home at any hour, compare prices, pay electronically and have goods delivered to their door, through e-shops and online banking. It saves time and travel, gives a wider choice than local shops, and lets even small businesses sell to customers across the country and the world.

11. How computer can be useful in weather forecasting?

For weather forecasting, computers collect accurate data from weather stations, satellites and radars all over the world, and then process this huge amount of data with mathematical models of the atmosphere. The results are forecast maps showing rain, wind, storms and temperature for the coming days. Because the calculations are enormous, only fast computers can produce forecasts in time to warn people about

severe weather.

12. Define the CBT training.

CBT (Computer Based Training) is training given through a computer instead of a classroom teacher. The learner works through lessons, exercises, pictures and tests on the computer, at his own speed and whenever he has time. CBT is used to train employees, teach driving, and give technical courses, because one program can train many people in different places at the same cost.

13. Describe the online shopping and banking.

Online shopping means buying goods through websites: the customer selects items in an e-shop, pays by card or electronic transfer, and the goods are delivered to his home.

Online banking means using the internet or an ATM to use bank services, such as checking balances, transferring money and paying bills, without visiting a branch. Both save time, work day and night, and need care about passwords and security.

14. Define the Videoconferencing.

Videoconferencing is a form of conferencing in which people in different places can see and hear each other at the same time over a network, using video cameras, microphones, speakers and screens. It is an advanced form of teleconferencing. It saves travel time and cost, and it is used for business meetings, distance teaching, medical consultation and interviews.

Key notes (this chapter):

- Chapter 4, MCQ (vii): the option is printed as 'nino second' and the printed key lists (d); the chapter text (p. 68) states that a computer works 'in a fraction of a second like nano second or even pico second', so the answer given here is (c) nano second. Chapter 4 has no question 4 in print: the numbering jumps from Q3 to Q5.

Chapter 5: Computer Architecture (Book pp. 71-90, Exercise 5C pp. 88-90)

Inside the CPU



Exercise 5C - 1. Fill in the blanks

(i) DMA stands for _____.

Answer: **Direct Memory Access**

(ii) In _____ mode, data can be transmitted in both directions simultaneously.

Answer: **Full Duplex**

(iii) The lexical analyzer is also commonly known as a _____.

Answer: **Lexer**

(iv) The _____ reads the instructions from the memory and decodes these instructions.

Answer: **Control Unit**

(v) The _____ interface consists of Window, Menus, Icons and pointers.

Answer: **Graphical User**

(vi) EEPROM stands for _____.

Answer: **Electrically Erasable Programmable Read Only Memory**

(vii) Stack pointer register is used for _____.

Answer: **Maintaining Stack**

(viii) DRAMs require _____ refreshing to maintain data storage.

Answer: **Periodic**

(ix) _____ language was developed for business applications.

Answer: **COBOL**

(x) _____ register is used in mathematical or logical operations.

Answer: **General Purpose**

(xi) DVD stands for _____.

Answer: **Digital Versatile Disk**

(xii) BIOS stands for _____.

Answer: **Basic Input/Output System**

(xiii) Initial work on the internet was done in _____ operating system.

Answer: **Unix**

(xiv) The instructions that are used to transfer data from one unit to another during program execution are called _____.

Answer: **Read/Write Instruction**

(xv) _____ registers are used in mathematical and logical operations.

Answer: **General Purpose**

Exercise 5C - 2. Choose the correct option

(i) Data and program not being used by computer are stored in:

a) Secondary storage ✓

b) cache

c) Primary storage

d) printer

(ii) A set of instructions that run the computer are:

a) hardware

b) document

c) CPUs

d) software ✓

(iii) The program that contains instructions to operate a device is called:

a) Device driver ✓

b) Device operator

c) Device linking

d) Device system

(iv) CPU is an example of:

a) software

b) A program

c) hardware ✓

d) an output unit

(v) The address of instruction under the processor execution is contained within:

a) Program Counter ✓

b) Current Instruction register

c) Memory Address register

d) Memory Buffer register

(vi) A computer derives its basic strength from:

a) Speed

b) Memory

c) Accuracy

d) All of the above ✓

(vii) The arithmetic/logic unit performs the following actions:

a) Control computer operations

b) Perform arithmetic functions such as addition and subtraction etc.

c) Perform logical comparisons such as equal, greater than, less than

d) Both b and c ✓

(viii) Which is a storage device:

a) CPU

b) Clock

c) Floppy disk ✓

d) Bus

(ix) Which component is responsible for comparing the contents of two pieces of data:

a) ALU ✓

b) Control unit

c) Memory

d) None

(x) Which one is faster:

a) RAM

b) Cache

c) Register ✓

d) Hard disk

Exercise 5C - 3. Write T for true and F for false statement

(i) Bps stands for byte per second.

Answer: True

(ii) In simplex transmission mode, communication can take place in both directions.

Answer: False

(iii) Random access memory is volatile memory.

Answer: True

(iv) Operating system is an application program.

Answer: False

(v) External buses and internal buses are similar.

Answer: False

(vi) Accumulator register is used to control the stacks in the computer.

Answer: False

(vii) LIFO stands for Last-in-first-off.

Answer: False

(viii) Expansion slot is a place where an expression card is fitted.

Answer: False

(ix) Static RAM holds the data as long as power is supplied to it.

Answer: True

(x) The clock of the computer ticks once in one second just like an ordinary clock.

Answer: False

(xi) VDU is an input device.

Answer: False

Exercise 5C - Short questions

4. What is CPU? Describe briefly.

The CPU (Central Processing Unit) is the brain of the computer: the hardware unit that carries out instructions and controls the whole system. It has three main parts: the Control Unit, which fetches and decodes instructions; the ALU (Arithmetic and Logic Unit), which performs calculations and comparisons; and the registers, small high-speed storage places inside the CPU that hold data and addresses during processing. The CPU works together with memory through buses.

5. Explain the architecture of computer system.

The architecture of a computer system is the organisation of its main parts and how they are connected. The CPU (Control Unit, ALU and registers) is joined by buses to the main memory (RAM and ROM) and to input and output devices through ports and controllers. Data and instructions travel on the data bus, addresses on the address bus, and control signals on the control bus. Secondary storage, such as hard disks, holds data and

programs when they are not in use.

6. Differentiate between the following: (i) PROM and EPROM (ii) Address Bus and Control Bus (iii) Serial and Parallel Ports (iv) Linker and Loader (v) CU and ALU.

(i) PROM (Programmable Read Only Memory) can be programmed once and cannot be changed; EPROM (Erasable Programmable Read Only Memory) can be erased with ultraviolet light and programmed again. (ii) The address bus carries the address of the memory location to be used; the control bus carries control signals such as read and write. (iii) A serial port sends data one bit at a time; a parallel port sends several bits at once over separate lines. (iv) The linker joins program parts into one executable program; the loader places that program into memory for running. (v) The CU (Control Unit) directs and controls all operations; the ALU performs arithmetic and logical operations.

7. Differentiate between Compiler and Interpreter.

A compiler translates the whole source program into machine code at once, producing an executable file; it is fast at run time, but errors are reported only after the whole translation. An interpreter translates and runs the program one statement at a time, stopping at the first error; it is slower at run time but easier for finding and correcting errors while learning and testing a program.

8. How to transfer data from CPU to memory? Explain in steps.

Data are transferred from the CPU to memory in these steps: (1) The CPU places the address of the memory location on the address bus. (2) It places the data to be stored on the data bus. (3) It sends a write signal on the control bus. (4) The memory unit reads the address and the write signal, and stores the data from the data bus into that location. (5) The memory sends back an acknowledgement, and the CPU continues with the next instruction.

9. Define the different types of RAM.

RAM (Random Access Memory) is the volatile main memory that holds data and programs while the computer is working. Its types are: DRAM (Dynamic RAM), which stores each bit as a charge that must be refreshed periodically; SRAM (Static RAM), which holds data as long as power is supplied and is faster but more expensive; and SDRAM (Synchronous DRAM), which works in step with the system clock and is used as the main memory of modern computers.

10. Define the general-purpose registers.

General-purpose registers are small, fast storage places inside the CPU that can hold data, addresses or results for any purpose during processing. Unlike special registers such as the program counter or accumulator, they are not fixed to one job: the programmer and the CPU use them freely for arithmetic, logical operations and temporary storage. Because they are inside the CPU, they are much faster than main memory.

11. Describe the Bus and its types.

A bus is a set of wires or lines that carries data, addresses and control signals between the parts of a computer. Its types are: the data bus, which carries the actual data; the address bus, which carries the address of the memory location or device; and the control bus, which carries control signals such as read, write and clock. Buses may be internal, inside the CPU, or external, joining the CPU to memory and devices.

12. Define the machine instructions.

Machine instructions are the binary-coded orders that the CPU can directly understand and execute. Each instruction has two parts: an operation code (opcode), which tells the CPU what operation to do, such as add, move or compare; and an operand, which gives the data or the address of the data to be used. Programs written in any language are finally translated into machine instructions before the CPU runs them.

13. Differentiate between Fetch instruction and Decode instruction.

The fetch step and the decode step are the first two stages of the CPU's instruction cycle. In the fetch step, the Control Unit takes the next instruction from memory, using the address in the program counter, and brings it into the instruction register. In the decode step, the Control Unit interprets (decodes) that instruction to find out which operation it orders and which data it needs, before the execute step carries it out.

14. Differentiate between ROM and RAM.

ROM (Read Only Memory) is non-volatile memory: it keeps its contents, such as the BIOS start-up program, even when the power is off, and it is normally only read, not written. RAM (Random Access Memory) is volatile memory: it holds the programs and data in current use, it can be both read and written, but its contents are lost when the power is switched off. ROM is small and permanent; RAM is large and temporary.

Key notes (this chapter):

- Chapter 5, True/False (ix): the statement 'Static RAM holds the data as long as power is supplied to it' is True (SRAM is volatile and keeps data only while powered); the book's printed key lists it as False, which appears to be a key error. The answer given here is True.

Chapter 6: Security, Copyright and the Law (Book pp. 91-102, Exercise 6C pp. 100-102)

Exercise 6C - 1. Fill in the blanks

(i) Making illegal copies of copyrighted software is called _____.

Answer: **piracy**

(ii) A special program that can detect and remove viruses from computer is called _____.

Answer: **anti-virus**

(iii) Software that is available free for a limited period is called _____.

Answer: **shareware**

(iv) When the virus starts to impact on data, it is known as _____.

Answer: **virus attack**

(v) IR stands for _____.

Answer: **input request**

(vi) _____ is a software used for data compression.

Answer: **winzip**

(vii) The right to use the software on the computer is called _____.

Answer: **licence**

(viii) Software is a _____ of person who developed it.

Answer: **property**

Exercise 6C - 2. Choose the correct option

(i) A virus program is usually hidden in:

a) The operating system only

b) An application program only

c) The disk drive

d) The operating system or application programs ✓

(ii) Most computer crimes are committed by:

a) Hackers

b) International spies

c) Highly trained computer consultants ✓

d) Company insiders who have no extraordinary technical ingenuity

(iii) Types of software that can be freely distributed without violating copyright laws are called:

a) Shareware

b) Public domain ✓

c) Copy protected

d) a and b

(iv) Information is:

a) A marketable commodity

b) Can be stolen while leaving the original behind

c) Should be free, according to the original hacker ethic

d) All of the above ✓

(v) A virus that replicates itself is called a:

a) Bug

b) Worm ✓

c) Vaccine

d) Bomb

(vi) Another name for free software:

a) Encrypted software

b) Copy protected software

c) Public domain software ✓

d) Shareware

(vii) Another name for anti virus is:

a) Vaccine ✓

b) Worm

c) Trojan horse

d) DES

(viii) Security protection for personal computers include:

a) Internal components

b) Locks and cables

c) Software

d) All of these ✓

(ix) A secret word or numbers to be typed in on a keyboard before an activity can take place are called:

a) Biometric data

b) Data encryption

c) Password ✓

d) Private word

(x) What is the most common computer crime of these listed below:

a) Extortion of bank funds

b) IRS database sabotage

c) Putting people on junk mailing lists ✓

d) Software piracy

Exercise 6C - 3. Write T for true and F for false statement

(i) Software error can result in data loss.

Answer: True

(ii) Any person can change password.

Answer: False

(iii) All viruses activate in exactly the same manner.

Answer: False

(iv) A full backup means that once a week you can perform a complete backup.

Answer: False

(v) IR stands for intellectual rights.

Answer: False

(vi) A computer virus is a part of hardware.

Answer: False

(vii) Passwords, auditor checks and separation of employee functions are data protection techniques.

Answer: True

(viii) No one has ever been able to read encrypted messages without key.

Answer: False

(ix) It is legitimate to make a copy of software for backup purpose.

Answer: True

(x) The computer fraud and abuse act of 1984 defines software piracy as crime.

Answer: True

Exercise 6C - Short questions

4. What is computer virus?

A computer virus is a harmful program, written by a person, that attaches itself to other programs or files and spreads from one computer to another, usually without the user's knowledge. When the infected program runs, the virus also runs, copies itself, and may damage data, slow down the system or stop it from working. Viruses spread through infected disks, downloads, e-mail attachments and networks.

5. Define the anti-virus software.

Anti-virus software, also called a vaccine, is a special program that detects, prevents and removes viruses from a computer. It scans files, disks and incoming e-mail, compares programs with a list of known viruses, and cleans or deletes infected files. To stay effective, anti-virus software must be updated regularly, because new viruses appear all the time.

6. How viruses may damage computer system?

Viruses damage a computer system by corrupting or deleting files and programs, changing data, filling up memory and disk space, and slowing down or stopping the system. Some viruses display annoying messages, while others steal or destroy important information and make the computer unusable. The damage begins when the virus activates, which may be at once or when a particular date or action triggers it.

7. Define Data protection and Piracy acts.

Data protection means the rules and techniques, such as passwords, audit checks and separation of duties, that keep data safe from loss, misuse and unauthorised access.

Piracy acts are the laws that make it a crime to make or use illegal copies of copyrighted software. For example, the Computer Fraud and Abuse Act of 1984 defines software piracy as a crime, and copyright law protects software as the property of the person who

developed it.

8. Describe the Legislation and Copyright Issues.

Legislation and copyright issues concern the legal protection of software and data. Software is the property of its developer, and a licence gives the buyer only the right to use it, not to copy or sell it. Making illegal copies is piracy and is punishable by law. Copyright law also covers books, music and films, and special laws, such as the Computer Fraud and Abuse Act of 1984, define computer crimes and their punishments.

9. Define the types of viruses.

The main types of viruses are: the boot sector virus, which infects the start-up part of a disk; the file virus, which attaches itself to program files; the worm, which copies and replicates itself across a network; the Trojan horse, which hides inside a harmless-looking program; the logic bomb, which activates on a particular event or date; and the macro virus, which spreads through documents such as word-processor files.

10. What is a password?

A password is a secret word or set of numbers that a user must type on the keyboard before an activity, such as logging into a computer or an account, can take place. It proves the user's identity and stops unauthorised persons from using the system or reading private data. A good password is kept secret, is hard to guess, and is changed regularly.

11. Write the names and define briefly the antivirus.

Common anti-virus programs (vaccines) are used to find and remove viruses: they scan the whole system, check new files and e-mail, and clean or quarantine infected items. Examples named in the chapter's era include Norton AntiVirus and McAfee VirusScan. Whichever program is used, it must be updated often with new virus definitions, because it can only recognise the viruses it knows.

12. Describe the exemption of 1990 act.

The exemption of the 1990 act refers to the legal exception that allows a lawful owner of software to make one copy of it for backup purposes. If the original disk is damaged or lost, the backup copy may be used instead, but it may not be sold, lent or given to others. This exemption protects the buyer against loss while still protecting the developer's copyright.

Key notes (this chapter):

- Chapter 6, Fill-in (v): the book's printed key gives 'input request' for 'IR stands for ___'; that expansion is not defined in the chapter text, so it is reproduced as printed and should be treated with caution.

Chapter 7: Windows Operating System (Book pp. 103-114, Exercise 7C pp. 113-114)

Exercise 7C - 1. Fill in the blanks

(i) GUI stands for _____.

Answer: **Graphical User Interface**

(ii) The capability of an operating system to load multiple programs into memory at one time is called _____.

Answer: **Multitasking**

(iii) With plug and play a computer can automatically _____ and _____ a device.

Answer: **Detect, Configure**

(iv) Multiple actions can be performed on a _____ file type. _____ partition refers to a portion of a disk that can contain other partitions.

Answer: **Registered, Extended**

(v) A _____ operating system allows multiple users to perform multiple tasks at the same time.

Answer: **Multi-User**

(vi) Windows explorer offers many actions that can be performed on _____ and _____.

Answer: **Files, Folders**

(vii) Windows maintains a _____ for all printing jobs.

Answer: **Print queue**

(viii) All deleted items are stored in _____.

Answer: **Recycle bin**

(ix) The Disk Management utility gives you a _____ interface.

Answer: **Graphical**

Exercise 7C - 2. Choose the correct option

(i) An operating system is a:

a) System Utility

b) Application Software

c) System software ✓

d) Software package

(ii) Ctrl + Alt + Del is:

a) An invalid key combination

b) Recognized by windows only ✓

c) Used to close the active window

d) Both b and c

(iii) As compared to command line operating system, a GUI operating system is:

a) More efficient

b) Easier to use ✓

c) More reliable

d) All of the above

(iv) The maximum number of primary partitions that can be created on a disk is:

a) Two

b) Three

c) Four ✓

d) None of the above

(v) Windows explorer is used to:

a) Access the Internet

b) Explore system resources

c) Perform maintenance on the hard disk

d) Navigate files and folders on the computer ✓

Exercise 7C - 3. Write T for true and F for false statement

(i) Primary partitions can not be created on basic disks.

Answer: False

(ii) Microsoft Windows is a single user operating system.

Answer: False

(iii) In windows explorer, left pane displays folders and drives on your computer in a tree view shape.

Answer: True

(iv) With Windows explorer, you can only manage the local files, folders and drives.

Answer: False

(v) GUI was first introduced by Apple's Macintosh computers.

Answer: True

(vi) Operating system is responsible for the effective use of computer system.

Answer: True

(vii) Maximum four primary partitions can be created on a basic disk.

Answer: True

(viii) Windows checks the file extension against a database of registered file types.

Answer: True

(ix) Disk management also indicates which drive contains the system partition.

Answer: True

(x) Prior to Windows installation on a computer, the disk is partitioned.

Answer: True

Exercise 7C - Short questions

4. Define operating system and discuss their types.

An operating system is the system software that manages the computer's hardware and lets the user run programs. Its types are: single-user operating systems, used by one person at a time; multi-user operating systems, which let many users work at the same time; multitasking operating systems, which run several programs at once; multiprocessing operating systems, which use more than one processor; and command-line and graphical (GUI) operating systems, which differ in how the user gives commands.

5. Give a comparison between Command Line operating system and Graphical User Interface.

A command line operating system takes orders typed as text commands; it is fast and uses little memory, but the user must remember commands and typing mistakes cause errors. A graphical user interface (GUI) gives orders by clicking icons, menus and windows

with a mouse; it is easier to learn and use, shows pictures and colours, but it needs more memory and a faster computer. Modern systems such as Windows use a GUI.

6. Discuss different features of Windows 2000 operating system.

Windows 2000 is a graphical operating system with a desktop, icons, windows, menus and a taskbar. Its features are: multitasking and multi-user working; plug and play, which detects and configures new devices automatically; Windows Explorer for managing files and folders; Disk Management for partitioning disks; a print queue for printing jobs; a Recycle Bin for deleted items; and network support for sharing files and printers.

7. Write a short note on each of the following: Disk Management Utility, Windows Explorer, Print Queue.

Disk Management Utility: a Windows tool that shows disks and partitions with a graphical interface, lets the user create and format partitions, and indicates which drive holds the system partition. Windows Explorer: a tool that displays folders and drives in a tree view and is used to navigate, copy, move, rename and delete files and folders. Print Queue: the waiting list Windows maintains for all printing jobs, so that documents are printed one after another in order.

8. What do you mean by Plug and Play? Does Windows 2000 provide this feature?

Plug and Play means that when a new device is connected, the computer automatically detects it and configures it, without the user having to set switches or install drivers by hand. Yes, Windows 2000 provides Plug and Play: on detecting a new device it finds the right driver, assigns resources to it, and makes it ready for use.

9. Define Partitioning. Briefly describe primary and extended partitioning.

Partitioning means dividing a hard disk into separate parts, called partitions, so that each part can be used as if it were a separate disk. A primary partition is a main partition that can hold an operating system and be booted from; at most four primary partitions can be created on a basic disk. An extended partition is a portion of a disk that can contain other (logical) partitions, so that more than four drives can be made on one disk.

10. Differentiate the following: Multitasking and Multiprocessing; File Management and Disk Management; Single-User operating system and Multi-User operating system.

Multitasking means one processor runs several programs by switching quickly between them; multiprocessing means two or more processors work together at the same time. File Management organises files and folders and controls access to them; Disk

Management organises the disk itself, its partitions and formatting. A single-user operating system serves one user at a time; a multi-user operating system lets several users work on the same computer at the same time.

Key notes (this chapter):

- Chapter 7, Fill-in (iv) carries two blanks (Registered file type; Extended partition), which is why the printed key lists ten entries for nine numbered items. Chapter 7, MCQ has only five items (i)-(v) in print.

Chapter 8: Word Processing (Book pp. 115-132, Exercise 8C pp. 130-132)

Exercise 8C - 1. Fill in the blanks

(i) The bar which contains the name of active application is known as _____.

Answer: **Title Bar**

(ii) WYSIWYG stands for _____.

Answer: **What You See Is What You Get**

(iii) The appearance or shape of a character is referred to as _____.

Answer: **Font**

(iv) The page orientation may be _____ or _____.

Answer: **Portrait, Landscape**

(v) _____ graphics can be thought of as painted pictures.

Answer: **Bitmap**

(vi) The _____ automatically moves to the next line when you have filled one line with text.

Answer: **Word wrap**

(vii) A _____ is a character or word that represents a series of keystrokes.

Answer: **Macro**

(viii) A built-in _____ allows you to search for synonyms.

Answer: **Thesaurus**

(ix) A _____ shows you the positioning of text, tabs, margins, indents and other elements

on the page.

Answer: **Ruler**

(x) In case of _____ mode the newly entered text is written over the existing text.

Answer: **Overtype**

Exercise 8C - 2. Choose the correct option

(i) Which of the following is a word processor:

a) Adobe Acrobat

b) Photo Express

c) MS Excel

d) MS Word ✓

(ii) Which of the following keyboard shortcuts is used to change the case?

a) Ctrl + F3

b) Shift + F3 ✓

c) Alt + F3

d) Ctrl + Shift + F3

(iii) In MS Word, the data that is being copied or moved is:

a) Temporarily stored in Recycle bin

b) Permanently stored in Recycle bin

c) Temporarily stored in Clipboard ✓

d) Permanently stored in Clipboard

(iv) Which of the following can be used to launch the Word Art:

a) Status bar

b) Ruler

c) Standard toolbar

d) Drawing toolbar ✓

(v) Which of the following feature enables you to reverse the changes you have made to the document?

a) WYSIWYG

b) Redo

c) Undo ✓

d) GUI

Exercise 8C - 3. Write T for true and F for false statement

(i) Word processor is just an electronic typewriter.

Answer: False

(ii) The bar containing the drop down menus is called scroll bar.

Answer: False

(iii) Font face is shown on Formatting tool bar.

Answer: True

(iv) Footnote appears at the bottom of every page.

Answer: False

(v) The interface represents the way through which you can interact with the word processing software.

Answer: True

(vi) In insertion mode the newly entered text is placed at the current position of the cursor.

Answer: True

(vii) Sans-Serif fonts have extra decorative lines at the ends of the strokes that make up each character.

Answer: False

(viii) Alignment refers to the orientation of the lines of a paragraph with respect to the margins.

Answer: True

(ix) Line Spacing refers to amount of space between paragraphs.

Answer: False

(x) Clipboard is managed by Microsoft Word.

Answer: False

Exercise 8C - Short questions

4. Write a note on the following: Word processor, Clipboard, Word Art.

Word processor: an application software that creates, edits, formats and prints text documents, with tools for fonts, margins, tables, pictures and spelling check; MS Word is an example. Clipboard: a temporary holding place in the computer's memory where data that is cut or copied is kept, so that it can be pasted into the same or another document.

Word Art: a feature that creates decorative, coloured and shaped text for headings and titles, launched from the Drawing toolbar.

5. What is a text editor? Describe its basic features.

A text editor is a simple program used to create and edit plain text files without formatting such as fonts and pictures. Its basic features are: typing and deleting text; moving the cursor and scrolling; cut, copy and paste; find and replace; and saving and opening text files. Notepad is a common example. A text editor is used for notes and for writing program code.

6. Describe features of a full-featured word processor.

A full-featured word processor provides: WYSIWYG display, so the screen shows the document as it will print; fonts, sizes, colours and styles; paragraph formatting, alignment, line spacing and indents; headers, footers, footnotes and page numbers; tables, pictures, Word Art and charts; spelling and grammar check with a thesaurus; mail merge; and print preview with page setup for portrait or landscape pages.

7. Define font and discuss its types.

A font is the appearance or shape of a set of characters: its design, size and style. Font types are: serif fonts, which have small decorative lines at the ends of the strokes; sans-serif fonts, which have no such lines and look plain; and decorative or script fonts, which imitate handwriting or display lettering. Fonts are also described by size in points and by styles such as bold, italic and underline.

8. Describe formatting features for paragraph formatting.

Paragraph formatting controls how a paragraph looks on the page. Its features are: alignment (left, right, centre or justified with respect to the margins); line spacing (the space between lines); paragraph spacing (the space before and after a paragraph); indents (the distance of the paragraph from the left and right margins, including first-line indent); and tabs, which set the positions where text stops across the line, as shown on the ruler.

9. Draw the mark sheet showing your name, father name, school name, year of passing SSC examination, date of birth, list of all subjects and their respective marks, total marks, percentage and the overall grade. [Practical task]

Practical task: draw a table with the given headings and fill in your own details. Specimen layout: a heading 'MARKS SHEET', then rows for Name, Father's Name, School Name, Year

of Passing SSC, and Date of Birth, followed by a table with columns Subject, Marks Obtained and Total Marks for every subject, and a final row for Total Marks, Percentage (obtained marks divided by total marks, multiplied by 100) and Overall Grade. The student's own real details must be entered in the table.

10. Arrange the text given in the following figure according to the mentioned format. [Text wrapping around an object]

Practical task: this is the 'text wrapping' (Text Around) feature of a word processor. To arrange text around a picture or box: insert the object, select it, open the Format menu and choose the wrapping style (for example Square or Tight), and place the object where it is wanted. The text then flows around the object instead of stopping above and below it, as shown in the book's figure headed 'The Text Around'.

Key notes (this chapter):

- Chapter 8 prints its True/False section (Q3) before its MCQ section (Q2); these notes present the MCQs first and then True/False, in numerical order, with the same questions and answers.

Chapter 9: Spread Sheet (Book pp. 133-146, Exercise 9C pp. 144-146)

Exercise 9C - 1. Fill in the blanks

(i) For the custom format ##.000, the number 242.59 will appear as _____.

Answer: **242.590**

(ii) The intersection of a row and a column forms a cell.

Answer: **Cell**

(iii) In a spreadsheet _____ are built in formulas.

Answer: **Functions**

(iv) Labels are used to identify a value or a series of values.

Answer: **Label**

(v) _____ are names that you define to represent a cell or cell range on a worksheet.

Answer: **Named Ranges**

(vi) Calling cells by just their addresses (such as 'A1') is called relative referencing.

Answer: **Relative Referencing**

(vii) An _____ is indicated by a bold rectangular border.

Answer: **Active Cell**

(viii) _____ is a software for manipulating numbers.

Answer: **Spreadsheet**

(ix) A _____ may contain multiple _____.

Answer: **Workbook, Worksheet**

(x) 0 is a placeholder that determines how many digits to display on either side of a decimal number.

Answer: **0**

Exercise 9C - 2. Choose the correct option

(i) Which of the following is a spreadsheet?

a) MS Word

b) MS Excel ✓

c) MS Powerpoint

d) Both b and a

(ii) The actual working area in Microsoft Excel is:

a) Workbook

b) Worksheet ✓

c) Spreadsheet

d) Note sheet

(iii) Which of the following is an absolute address?

a) A1

b) A1$

c) A1 ✓

d) None of the above

(iv) Formula can only be applied on:

a) Values ✓

b) Labels

c) Unmerged cells

d) None of the above

(v) Which of the following function is used to get the current date?

a) Exact()

b) Today() ✓

c) Month()

d) Year()

Exercise 9C - 3. Write T for true and F for false statement

(i) Because Microsoft Excel is a spreadsheet, therefore it does not have a spellchecker component.

Answer: False

(ii) Functions can be a more efficient way of performing mathematical operations than defining your own formulas.

Answer: True

(iii) A formula can not manipulate labels.

Answer: True

(iv) Worksheet is the basic unit where the data is manipulated in a workbook.

Answer: False

(v) Footnote can not be applied in a spreadsheet software.

Answer: False

(vi) By default, the numbers as well as the text is aligned RIGHT in a cell.

Answer: False

(vii) A formula containing relative referencing is not copied exactly.

Answer: True

(viii) # is a place holder similar to the 0 character, except that space is left for insignificant zero characters on either side of a decimal point.

Answer: False

(ix) A formula containing absolute referencing is not copied exactly.

Answer: False

(x) In MS Excel, a worksheet can have maximum 65,536 rows.

Answer: True

Exercise 9C - Short questions

4. Define spreadsheet and discuss its basic features.

A spreadsheet is a software for manipulating numbers, arranged in a grid of rows and columns whose intersections form cells. Its basic features are: entering labels (text) and values (numbers); built-in formulas and functions for calculations; relative and absolute cell referencing; formatting of numbers, fonts and borders; charts such as pie charts; sorting and printing; and page setup with margins, headers and footers. MS Excel is the best-known spreadsheet.

5. Differentiate the following: Workbook and Worksheet; Active Cell and Passive Cell; Word Processor and Spreadsheet; Function and formula; Labels and Values.

Workbook and Worksheet: a workbook is a file that may contain many worksheets; a worksheet is one page (grid) inside it. Active Cell and Passive Cell: the active cell is the selected cell, shown by a bold border, ready for entry; other cells are passive. Word Processor and Spreadsheet: a word processor handles text documents, a spreadsheet handles numbers and calculations. Function and formula: a function is a built-in ready formula such as SUM(); a formula is an expression the user writes, such as =A1+B1. Labels and Values: labels are text that identify data; values are numbers on which calculations are done.

6. Every cell in a worksheet has a reference number, how is it calculated? Discuss with examples the major differences between Relative and Absolute referencing.

A cell's reference is its column letter followed by its row number, for example A1, B5 or C10. In relative referencing, the reference changes when a formula is copied: copying =A1+B1 from C1 to C2 gives =A2+B2, because the references move with the new position. In absolute referencing, dollar signs fix the reference, for example A1, so copying the formula anywhere still points to A1. Mixed referencing fixes only the row or only the column, such as A$1 or $A1.

7. Create a Pie chart for the marks obtained by your class fellows in SSC examination. [Practical task]

Practical task: enter your class fellows' names and their SSC marks in two columns of a worksheet, select the data, and use the Chart Wizard to insert a Pie chart, which shows each student's share of the total as a slice of a circle. Give the chart a title, label each slice with the student's name, and check that the slices add up to the whole class. Your own class data must be used.

8. What do you understand by the term name ranges? Can it be helpful in simplifying the

worksheet?

Name ranges (named ranges) are names that the user defines to represent a cell or a range of cells on a worksheet, for example naming B2:B10 as 'Marks'. Yes, they simplify the worksheet: formulas become easy to read, such as =SUM(Marks) instead of =SUM(B2:B10), and mistakes are reduced, because the name stays correct even if rows or columns are added inside the range.

9. What are the advantages of using a spreadsheet program?

A spreadsheet program is fast and accurate: it recalculates all results at once when a number changes, so there is no need to redo long calculations by hand. It organises data in rows and columns, applies the same formula to many cells, draws charts from data, sorts and filters records, and prints neat tables. These advantages make it useful for accounts, results, budgets, stock and scientific data.

10. Insert another worksheet in the workbook of Q.4 to develop a Pie chart for the marks obtained by the students of your neighboring section in SSC examination. Calculate overall pass percentage, and average marks of the students of both sections in another (third) worksheet. [Practical task]

Practical task: keep the first section's data and pie chart in Sheet1, enter the neighbouring section's data in Sheet2 and make its pie chart the same way, then in Sheet3 link the two sheets to calculate the combined results. Overall pass percentage = (total passed students in both sections divided by total students in both sections) multiplied by 100. Average marks = (sum of all marks in both sections) divided by (total number of students). Use formulas that pick the data from Sheet1 and Sheet2, so the results update if the marks change.

Chapter 10: Fundamentals of the Internet (Book pp. 147-154, Exercise 10C pp. 153-154)

Exercise 10C - 1. Fill in the blanks

(i) Collection of related web pages is called _____.

Answer: **Web Site**

(ii) Initially ARPANET was a _____ connecting small number of users.

Answer: **Wide Area Network**

(iii) News server use _____ protocol to transfer articles among them.

Answer: **NNTP**

(iv) NSF established a separate high speed network called _____.

Answer: **NSFnet**

(v) The LAN can be connected to ISP using a high-speed phone line called a _____.

Answer: **T1 line**

(vi) World Wide Web was established in _____.

Answer: **1989**

(vii) MIME stands for _____.

Answer: **Multipurpose Internet Mail Extension**

(viii) The four numbers in an IP address are called _____.

Answer: **octets**

(ix) _____ address is easy to remember.

Answer: **DNS**

(x) HTML stands for _____.

Answer: **Hyper Text Markup Language**

Exercise 10C - 2. Choose the correct option

(i) A computer can be linked to the Internet through:

a) A phone-line modem

b) DSL

c) Cable modem

d) All of the above ✓

(ii) Which of the following is an email client?

a) Internet Explorer

b) Outlook Express ✓

c) Google

d) None of the above

(iii) Which of the following protocol is used to access web pages on World Wide Web?

a) TCP/IP

b) Gopher

c) HTTP ✓

d) HTML

(iv) Which of the following is used to find information on World Wide Web?

a) Web Browser

b) Web Site

c) Search Engine ✓

d) Web Server

(v) The length of IP Address is:

a) 8 bit

b) 16 bit

c) 32 bit ✓

d) 64 bit

Exercise 10C - 3. Write T for true and F for false statement

(i) IP address is a 16-bit addressing scheme.

Answer: False

(ii) Every computer on a network has a Network Interface Card that directly connects it to the other computers.

Answer: True

(iii) URL is used to locate a computer on the Internet.

Answer: False

(iv) A hypertext document is also called web page.

Answer: True

(v) email server is a software used to create, send and receive emails.

Answer: False

(vi) Each computer on the Internet must have HTTP Server configured to connect to the Internet.

Answer: False

(vii) NNTP stands for National News Transmission Protocol.

Answer: False

(viii) On Internet, Junk mails are also called spam.

Answer: True

(ix) News Server is used to streamline the news transmission of the local radio stations.

Answer: False

(x) You can not attached a file larger than 20MB to an email.

Answer: False

Exercise 10C - Short questions

4. Briefly describe the history of the Internet.

The internet began as ARPANET, a wide area network set up for a small number of users to share research between computers. Later the NSF established a separate high-speed network called NSFnet, and other networks joined, so the system grew into a world-wide network of networks. In 1989 the World Wide Web was established, giving easy, linked access to documents, and the internet then spread to homes, schools, offices and businesses everywhere.

5. Write a note on World Wide Web.

The World Wide Web (WWW) is the system of linked web pages on the internet, established in 1989. A collection of related web pages is called a web site, held on a web server and viewed with a web browser. Pages are written in HTML (Hyper Text Markup Language) and are reached by their address (URL), and they are linked by hyperlinks, so a reader can jump from one page to another. The Web is used for information, education, business and entertainment.

6. What do we mean by Addressing schemes? How many types of addressing is used on the Internet? Discuss briefly.

Addressing schemes are the systems used to identify computers and resources on the internet so that data reach the right place. Two main types are used: IP addresses and DNS addresses. An IP address is a 32-bit numeric address, written as four numbers called octets, for example 192.168.1.1. A DNS (Domain Name System) address is the easy-to-remember name form, such as www.example.com, which the system translates into the matching IP address.

7. What do you know about Email? Discuss briefly.

E-mail (electronic mail) is the exchange of messages over the internet. To send and receive e-mail, a user needs an e-mail account on an e-mail server and an e-mail address, written as user name @ DNS address, for example name@example.com. Messages are composed in an e-mail client such as Outlook Express, sent through the server, and

stored in the receiver's mailbox until read. Files can be attached to messages, and unwanted junk mail is called spam.

8. Write a note on the following: Newsgroups, Search Engine, TCP/IP.

Newsgroups: discussion groups on the internet, classified by subject, where participants post messages for others to read and reply to; articles are exchanged between news servers using NNTP (Network News Transfer Protocol). Search Engine: a program, such as Google, that searches the Web for information on a topic and lists matching pages. TCP/IP: Transmission Control Protocol / Internet Protocol, the basic set of rules of the internet; IP addresses and routes the data, and TCP breaks it into packets, checks them and puts them back in order.

9. Discuss limitations on email.

E-mail has several limitations: the sender may not know the person he is communicating with, because messages are only text and can give a wrong impression, and some people misrepresent themselves. Messages can be delayed, lost or blocked, and mailboxes can be filled with spam. Attachments may carry viruses, private mail can be read if it is not protected, and e-mail cannot carry the tone of voice or expressions of face-to-face talk.

10. How the use of Internet is affecting our society? Give your comments.

The internet is changing society in good and bad ways. It gives instant access to information, education, jobs, shopping, banking and contact with people anywhere, and it helps businesses and governments work faster. On the other hand, it spreads false information, wastes time, weakens privacy, and can harm health and social life when used too much. Used wisely, with care for truth, safety and balance, the internet is a great benefit to society.